

1) Let $X = (X_1, X_2, X_3, \dots, X_n)$ and $Y = (Y_1, Y_2, Y_3, \dots, Y_n)$ be two linked lists. Write an implementation level function to merge the two lists together to obtain the linked list Z such that $Z = (X_1, Y_1, X_2, Y_2, \dots, X_n, Y_n)$. The two lists X, Y should be returned empty. (5 Marks)

2) What is the output of the following segment of code –Verify your answer with drawing (5 Marks)

```
Queue Q;           Stack S;           char C;
CreatQueue(&Q);      CreatStack(&S);
Push('t', &S);       Push('s', &S);
Push('e', &S);       Push('T', &S);
while(!IsEmpty(S)){
    Pop(&C, &S);      Enqueue(C, &Q);
}
while(!IsEmpty(Q))
{
    Dequeue(&C, &Q);  Printf("%c ", C);
}
```

3) Define a suitable data structure that can be used to store the data of the students in the faculty. The data are ordered and retrieved according to their arrival sequence. The total number of students cannot be expected, but it is required to be known frequently. **Write the declaration of the proposed structure and a function to add any new student data.** (5 Marks)

4) Knowing that we have a queue Q that has a set of characters, write a segment of code which uses a stack to print those characters in reverse order and keeps them in the queue. Assuming that you do not have any knowledge of how the stack and queue are implemented. (5 Marks)